

MCS-1: Probability and Statistics (MA6.302)

Monsoon 2026

Lecture 8

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August 28th, 2025



Multiple Random Variables

- So far we have been talking about single random variables and associated PMF's. However, often we are interested in multiple random variables.
- Consider two discrete random variables X , and Y associated with the same experiment.
- The **joint PMF** of X and Y are defined as

$$p_{X,Y}(x,y) = P(X = x, Y = y)$$

for all pairs of values x and y .

- This is none other than $P(\{X = x\} \cap \{Y = y\})$.
- Of course, the order does not matter.

Properties of Joint PMF

- Recall that if $A_1, A_2, \dots, A_K$ is a partition of sample space Ω ,

$$P(B) = P(B \cap \cup_{i=1}^K A_i) = P(\cup_{i=1}^K B \cap A_i) = \sum_{i=1}^K P(B \cap A_i)$$

- $\{X = x\}$ is the disjoint union of $\{X = x\} \cap \{Y = y\}$ for all y values Y can take.
- $\{X = x\} \cap \{Y = y\}$ is none other than $\{X = x, Y = y\}$.
- We can extend this to PMFs:

$$\sum_y P(X = x, Y = y) = P(X = x)$$

$$\sum_x P(X = x, Y = y) = P(Y = y)$$

- These are also called the marginal PMF's.
- So, $\sum_x \sum_y P(X = x, Y = y) = 1$.
- $\{X = x, Y = y\}$ for all pairs of numerical values taken by X and Y form a partition of the sample space.
- And now the normalization rule gives us the result!

Functions of Multiple Random Variables

- $\mathbb{E}(g(X, Y)) = \sum_x \sum_y g(x, y)P(X = x, Y = y).$
- Let $g(X, Y) = aX + bY$. Then,
 $\mathbb{E}(g(X, Y)) = \sum_x \sum_y (ax + by)P(X = x, Y = y) = a\mathbb{E}[X] + b\mathbb{E}[Y].$
- What if $g(X, Y) = aX^2 + bY^2 + c$?

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- What if $g(X, Y) = aX^2 + bY^2 + c$?
 $\mathbb{E}[g(X, Y)] = a\mathbb{E}[X^2] + b\mathbb{E}[Y^2] + c.$

Multiple Random Variables

- How about three random variables?
- We will write $p_{X,Y,Z}(x,y,z) = P(X = x, Y = y, Z = z)$.
- The rules are the same:

$$P(X = x, Y = y) = \sum_z P(X = x, Y = y, Z = z)$$

$$P(X = x) = \sum_y \sum_z P(X = x, Y = y, Z = z)$$

$$P(Y = y) = \sum_x \sum_z P(X = x, Y = y, Z = z)$$

$$P(Z = z) = \sum_x \sum_y P(X = x, Y = y, Z = z)$$

$$\sum_x \sum_y \sum_z P(X = x, Y = y, Z = z) = 1$$

- Generalizes easily to more than 3 random variables.

Linearity of Expectation

- Perhaps one of the most useful and powerful results!
- $\mathbb{E}[aX + bY + cZ + d] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c\mathbb{E}[Z] + d$
- More generally,
$$\mathbb{E}[a_1X_1 + a_2X_2 + \dots + a_nX_n] = a_1\mathbb{E}[X_1] + a_2\mathbb{E}[X_2] + \dots + a_n\mathbb{E}[X_n].$$
- This is extremely general! $X_1, \dots, X_n$ do not have to be mutually independent for this to hold!
- This generalizes to $\mathbb{E}[\sum_i a_i f_i(X_i)] = \sum_i a_i \mathbb{E}[f_i(X_i)]$, as long as the expectations are well defined.

Expectation of Binomial(n, p)

- Remember that a Binomial(n, p) random variable is nothing other than the sum of n independent Bernoulli's!
- $Y = \sum_{i=1}^n X_i$, where $X_i \sim \text{Bernoulli}(p)$.
- We know that $\mathbb{E}[X_i] = p$.
- Using our new found tool, we have:
$$\mathbb{E}[Y] = \mathbb{E}[\sum_{i=1}^n X_i] = \sum_{i=1}^n \mathbb{E}[X_i] = np.$$
- We do not need the mutual independence of the Bernoullis to get this result!

Balls and Bins

- I am throwing m distinguishable balls into n distinguishable bins. What is the expected number of empty bins (call this Y)? Every ball has to hit a bin and a bin can have multiple balls.
- Let

$$X_i = \begin{cases} 1, & \text{ith bin is empty} \\ 0, & \text{otherwise} \end{cases}$$

- We want $\mathbb{E}[Y] = \mathbb{E}[\sum_{i=1}^n X_i]$.
- $\mathbb{E}[X_i] = P(\text{No ball falls in bin } i) = (1 - \frac{1}{n})^m$
- $\mathbb{E}[Y] = n(1 - 1/n)^m$
- When $m = n$, for large n , $\mathbb{E}[Y] = n(1 - 1/n)^n \approx n/e$.

Conditioning One Random Variable Over Another

- Let X and Y be two random variables associated with the same experiment. Now the knowledge of $Y = y$ for some particular value y provides us with partial knowledge about what value X may take.
- The conditional PMF of X given Y is given by $p_{X|Y}(x, y) = P(X = x | \{Y = y\})$.
- Using the same set of rules as before we can write:

$$\begin{aligned} P(X = x | Y = y) &= \frac{P(X = x, Y = y)}{P(Y = y)} \\ &= \frac{p_{X,Y}(x, y)}{p_Y(y)} \end{aligned}$$

- For X any fixed y such that $P(Y = y) > 0$, we also have: $\sum_x P(X = x | Y = y) = 1$.
- So, a conditional PMF satisfies the properties of a PMF.

Example: Conditional PMF

	Right Handed ($L=0$)	Left handed ($L=1$)	
Men ($X=0$)	43	7	50
Women ($X=1$)	47	3	50
	90	10	100

- Bob and Alice are interested in finding out the conditional probability of being left handed given a person is a man. Bob finds his data again.
- $P(L = 1|X = 0)$ is just the fraction of all men who are left handed
- $P(L = 1|X = 0) = 7/50$.
- Let us plug in the formula. $P(L = 1, X = 0) = 7/100$.
- $P(X = 0) = 50/100$. So, $P(L = 1|X = 0) = \frac{P(L=1, X=0)}{P(X=0)} = 7/50$.
- What is $P(L = 0|X = 0)$? Its just the fraction of all men who are right handed! So $43/50$.
- $P(L = 0|X = 0) + P(L = 1|X = 0) = 1!$

Independence of Random Variables

- Lets first consider two events $\{X = x\}$ and A . We know that these two events are independent if $P(\{X = x\}, A) = P(\{X = x\})P(A)$
- In other words if $P(A) > 0$, then $P(X = x|A) = P(X = x)$, i.e. knowing the occurrence of A does not change our belief about $\{X = x\}$.
- We will call the random variable X and event A to be independent if $P(X = x, A) = P(X = x)P(A)$ For all x
- Two random variables are said to be independent if $P(X = x, Y = y) = P(X = x)P(Y = y)$ For all x and y
- To put it a bit differently, $P(X = x|Y = y) = P(X = x)$ For all x and y such that $P(Y = y) > 0$

An Important Implication

- We saw that $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$ no matter whether X and Y are independent or not.
- If X and Y are independent, $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$
- In fact, $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$

Variance of Sum of Random Variables

- Let X and Y be two independent random variables. What is $\text{var}(X + Y)$?
- Then, $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$.
- Variance of sum of independent random variables equals the sum of the variances!

Independence Of Several Random Variables

- Three random variables X , Y and Z are said to be independent if $P(X = x, Y = y, Z = z) = P(X = x)P(Y = y)P(Z = z)$ For all x, y, z
- If X, Y, Z are independent, then so are $f(X), g(Y)$ and $h(Z)$.
 - Also, any random variable $f(X, Y)$ and $g(Z)$ are independent.
 - Are $f(X, Y)$ and $g(Y, Z)$ independent? **Not necessarily, both have Y in common.**
- For n independent random variables, $X_1, X_2, \dots, X_n$, we also have:
$$\text{var}(X_1 + X_2 + X_3 + \dots + X_n) = \text{var}(X_1) + \text{var}(X_2) + \dots + \text{var}(X_n)$$

Variance of Binomial

- Consider n independent Bernoulli variables $X_1, X_2, \dots, X_n$, each with probability p of having value “1”. The sum $Y = \sum_{i=1}^n X_i$ is Binomial(n, p) random variable where X_i is a Bernoulli(p) random variable.
- We saw last time that $\mathbb{E}[Y] = \mathbb{E}[\sum_{i=1}^n X_i] = \sum_{i=1}^n \mathbb{E}[X_i] = np$.
- What about the variance?
- Recall that $\text{var}(X_i) = p(1 - p)$ for $i \in \{1, 2, \dots, n\}$.
- $\text{var}(Y) = \text{var}(X_1 + X_2 + \dots + X_n) = \sum_{i=1}^n \text{var}(X_i) = np(1 - p)$.

Conditional Independence

- Very similar to conditional independence of events!
- X and Y are conditionally independent, given a positive probability event A if $P(X = x, Y = y|A) = P(X = x|A)P(Y = y|A)$ For all x and y
- Same as saying $P(X = x|Y = y, A) = P(X = x|A)$, i.e., once you know that A has occurred, knowing $\{Y = y\}$ has occurred does not give you any more information!
- Like we learned before, conditional independence does not imply unconditional independence.

Continuous Random Variable

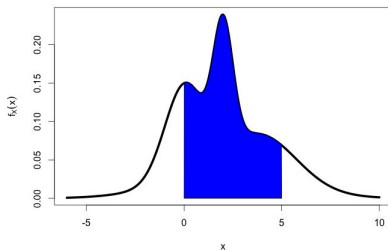
- A continuous random variable is a random variable that:
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- Examples might include:
 - The time at which a bus arrives.
 - The volume of water passing through a pipe over a given time period.
 - The height of a randomly selected individual.

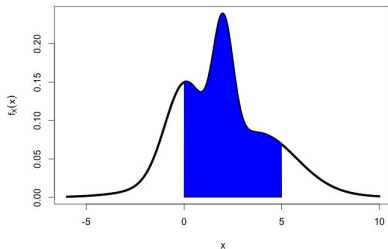
Probability Density Function

- For a continuous random variable, we cannot construct a PMF - each specific value has zero probability.



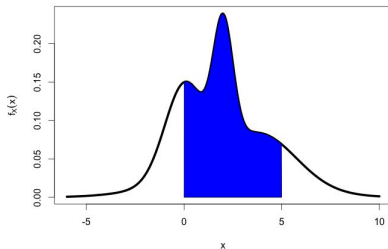
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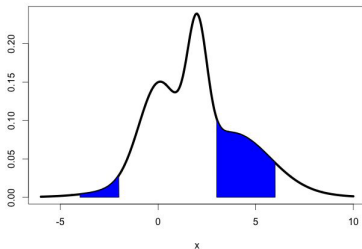


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- Instead, we use a continuous, non-negative function $f_X(x)$ called the probability density function, or PDF, of X .
- The probability of X lying between two values x_1 and x_2 is simply the area under the PDF, i.e. $P(a \leq X \leq b) = \int_a^b f_X(x)dx$



Probability Density Function



- More generally, for any subset B of the real line,
$$P(X \in B) = \int_B f_X(x) dx$$
- Here, $B = (-4, -2) \cup (3, 6)$.

- Note that $f_X(a)$ is not $P(X = a)$!

Properties of PDF

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- To have normalization, we require,
$$\int_{-\infty}^{\infty} f_X(x)dx = P(-\infty < X < \infty) = 1. \text{ Total area under curve is 1.}$$

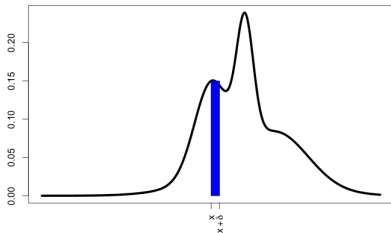
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$$\int_{-\infty}^{\infty} f_X(x)dx = P(-\infty < X < \infty) = 1. \text{ Total area under curve is 1.}$$
- Note that $f_X(x)$ can be greater than 1 – even infinite! – for certain values of x , provided the integral over all x is 1.

- We can think of the probability of our random variable lying in some small interval of length δ , $[x, x + \delta]$.

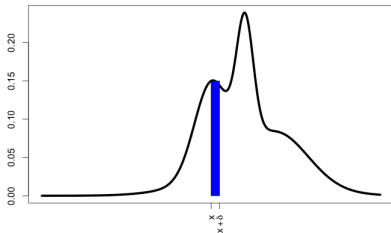
Intuition

- We can think of the probability of our random variable lying in some small interval of length δ , $[x, x + \delta]$.
- $P(X \in [x, x + \delta]) = \int_x^{x+\delta} f_X(t) dt \approx f_X(x)\delta$



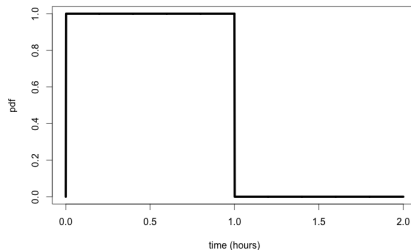
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- Note however that $f_X(x)$ is not the probability at x .

Continuous Uniform Random Variable



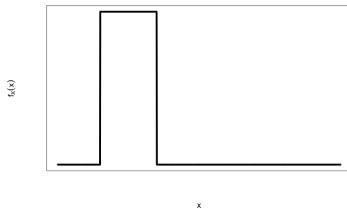
$$f_X(x) = \begin{cases} 1, & \text{if } 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

- What is $P(X > 0.6)$?
- What is $P(X > 1.1)$?
- What is $P(X = 0.7)$?

General Continuous Uniform Random Variable

- More generally, X is a continuous uniform random variable if it has PDF

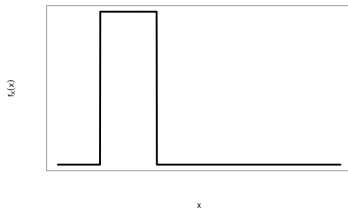
$$f(x) = \begin{cases} c, & \text{if } a \leq x \leq b \\ 0, & \text{else} \end{cases}$$



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$$f(x) = \begin{cases} c, & \text{if } a \leq x \leq b \\ 0, & \text{else} \end{cases}$$



- What is c ?
 - Well first let's see that $\int_a^b f_X(x) dx = c(b - a)$.
 - But, we want this to be 1. So c is $c = 1/(b - a)$.

Cumulative Distribution Function (CDF)

- Often we are interested in $P(X \leq x)$. For example,
 - What is the probability that the bus arrives before 1:30?
 - What is the probability that a randomly selected person is under 5'7"?
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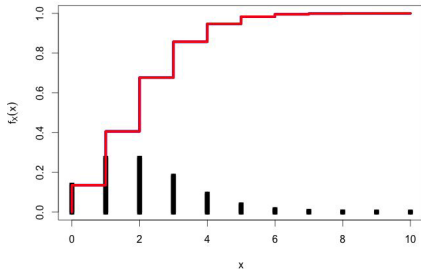
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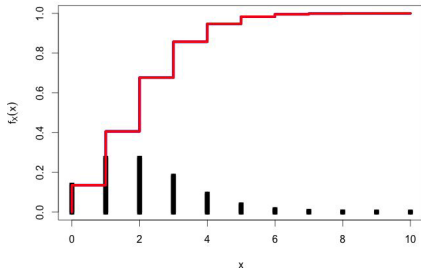
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- **Note:** If we know $P(X \leq x)$, we also know $P(X > x)$.

CDF of Discrete Random Variable



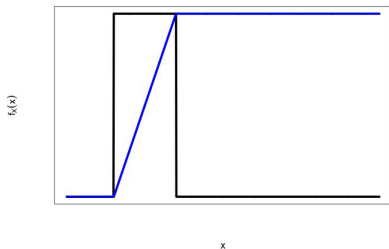
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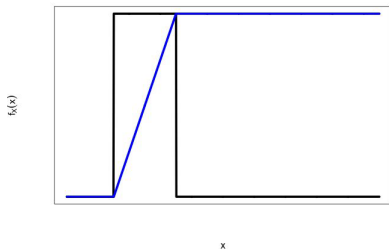
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Properties of CDF of Continuous Random Variable



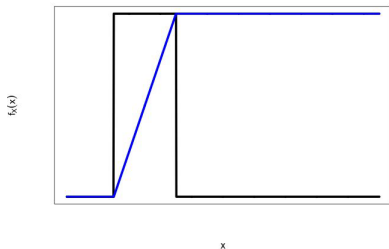
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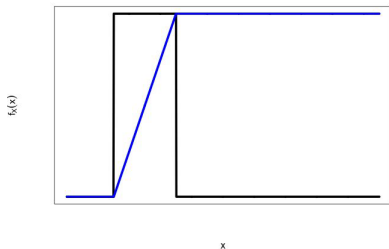
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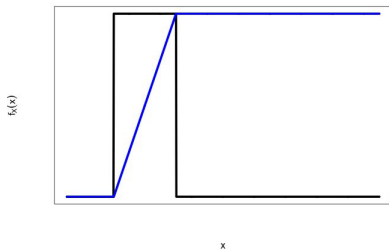
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- $F_X(x) \rightarrow 1$ as $x \rightarrow \infty$

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- What do you think the expectation of a continuous random variable is?
- $\mathbb{E}[X] = \int_{-\infty}^{\infty} xf_X(x)dx$
- Just as in the discrete case, we can think of $\mathbb{E}[X]$ as the “center of gravity” of the PDF.

Expectation of Function of Continuous Random Variable

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- Note that $g(X)$ can be a continuous random variable (e.g., $g(X) = X^2$) or $g(X)$ can be a discrete random variable (e.g., $g(X) = \text{sign}(X)$).

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- Then, the expectation of $g(X)$ is as follows.

$$\mathbb{E}[Y] = \int_{-\infty}^{\infty} g(x)f_X(x)dx$$

- Note that $g(X)$ can be a continuous random variable (e.g., $g(X) = X^2$) or $g(X)$ can be a discrete random variable (e.g., $g(X) = \text{sign}(X)$).
- Variance of X is $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

Mean of a Uniform Random Variable

- Let X be a uniform random variable over $[a, b]$. So, the pdf is

$$f_X(x) = \begin{cases} 0, & \text{if } x < a \\ \frac{1}{b-a}, & \text{if } a \leq x \leq b \\ 0, & \text{if } x > b \end{cases}$$

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- What is its expected value?

$$\begin{aligned} \mathbb{E}[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \frac{1}{b-a} \int_a^b x dx \\ &= \frac{1}{b-a} \frac{b^2 - a^2}{2} = \frac{a+b}{2} \end{aligned}$$

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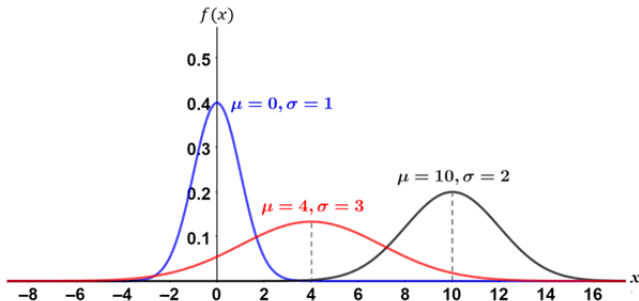
- Thus, $\text{Var}(X) = \frac{1}{3}(a^2 + ab + b^2) - \frac{1}{4}(a+b)^2 = \frac{(b-a)^2}{12}$

The Normal Distribution

- A normal, or Gaussian is a continuous random variable with pdf as follows.

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad x \in (-\infty, \infty)$$

where μ and $\sigma > 0$ are scalars.



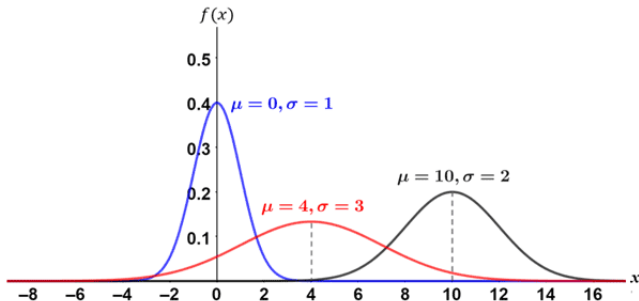
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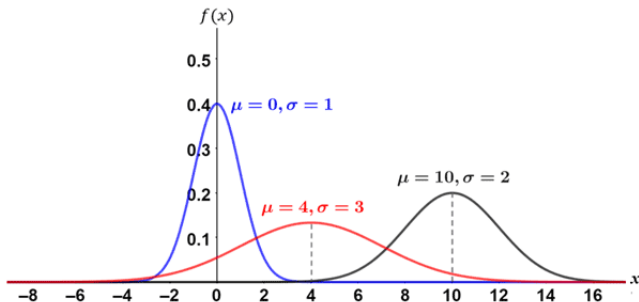
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- the mean of X is μ and variance is σ^2 . (proof next page)



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- By changing the variable using $y = x - \mu$

$$\mathbb{E}[X] = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} ye^{-\frac{y^2}{2\sigma^2}} dy + \mu = 0 + \mu$$

where the first integration is zero because the function $ye^{-\frac{y^2}{2\sigma^2}}$ is odd.

Variance of Gaussian Random Variable

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where we used the fact that $y^2 e^{-\frac{y^2}{2\sigma^2}}$ is even function of y .

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- Using the definition of Gamma function, $\Gamma(n) = \int_0^{\infty} z^{n-1} e^{-z} dz$

$$Var(X) = \frac{2\sigma^2}{\sqrt{\pi}} \Gamma\left(1 + \frac{1}{2}\right) = \frac{2\sigma^2}{\sqrt{\pi}} \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{2\sigma^2}{\sqrt{\pi}} \frac{1}{2} \sqrt{\pi} = \sigma^2$$

where we have used the fact that $\Gamma(n+1) = n \Gamma(n)$ and $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.